



SATHYA VARSHINI M

Embedded System Engineer

☎ 9629012848 ✉ sathyavarshini.13@gmail.com 🔗 <https://www.linkedin.com/in/sathya-varshini-m-8aa36724a>

CAREER OBJECTIVE

As an electronics engineering student, I'm looking for a job in the field of Embedded systems both in hardware and software. An individual who is passionate about simulation implementation design and digital circuits. I have an experience as part of a team. I have trained for 6 months in this field and completed a certification course in it. I have a strong desire to help the company succeed by creating innovative technological systems.

EDUCATION

2021 - 2025 VCET-Erode	ELECTRONICS AND COMMUNICATION ENGINEERING CGPA: 8.27
2020 - 2021 Jaycees Mt.Hr.Sec.School	HSC PERCENTAGE-90.20
2018 - 2019 Jaycees Mt.Hr.Sec.School	SSLC PERCENTAGE- 90.80

PROJECTS

IoT-Enabled baby cradle with advanced monitoring and alert system :

- The smart cradle baby monitoring system includes an Arduino UNO for controlling, ESP8266 module for wireless communication, a wet sensor for detecting moisture, a sound sensor for detecting the baby's cries, an MP3 module and speaker for soothing music, an LCD display for real-time data, a servo motor for swinging the cradle, DHT11 for temperature and humidity monitoring, MQ2 sensor for air quality monitoring, gyro sensor for fall detection and 4x2400 mAh battery for power backup.
- It incorporates a Blynk application for remote monitoring and receiving alerts.

Adaptive book holders and page turner for physically challenge:

- Designed an assistive device for differently-abled users to turn physical book pages using ESP32 microcontroller and 5 servo motors (2 for page holding, 2 for wheel rotation, 1 for page flipping).
- Built a Flutter mobile app with voice command integration for navigation (Next Page, Previous Page, Make Bookmark, Go to Bookmark).
- Programmed hardware control in Embedded C on ESP32 for synchronized servo operation.
- Implemented bookmark functionality with memory storage and automatic navigation to bookmarked pages.
- Enhanced accessibility by combining IoT, embedded systems, and mobile app development into a low-cost, user-friendly solution.

SKILLS

- **Design tools :** Mplab, keil uVision4 , Visual code, Arduino IDE, STM32Cube, Proteus
- **Hardware tools :** Microcontrollers- STM32, ARM (LPC2148), PIC (16F877A), Arduino UNO
- **Protocols :** UART, SPI, I2C
- **Languages :** C, Embedded C

CERTIFICATIONS

IPCS GLOBAL INSTITUTE- Embedded System Engineer

NPTEL-Principles of management

IIT Madras Research Park -7 Days Internship on Artificial Intelligence

Hackathon- Socio tech hackathon held at KPR college made by TamilNadu startup team

ACHIEVEMENT

State level Volleyball player(Khelo India)

DECLARATION

I hereby confirm that all the above information is true to the best of my knowledge, and I take full responsibility for any inaccuracies.